

Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: Old Data Only

Section 1 — Diagnostics

Period Definitions

```
pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
```

Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

Dataset Variant × Pandemic Period — messages and unique users
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

variant	period	messages	users
old	pre	4505	169
old	during	11102	291
old	post	2943	96
new_only	pre	94	15
new_only	during	318	23
new_only	post	25092	520
combined	pre	4865	177
combined	during	11960	307
combined	post	6592	247

Users by Number of Periods Appeared In
Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4

periods_appeared_in	n_users	pct_users
1	318	74.5
2	89	20.8
3	20	4.7

Section 2 — Dictionary Introspection

Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries
Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)

yahya_group	liwc2015	liwc22	dic_crosscheck
pronouns	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you
function_words	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant
other_grammar	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb
tense	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent
emotion	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad
informal	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear
summary	—	—	—
absolutist	absolutist	—	—

Section 3 — Main Results (all users)

Kruskal-Wallis Omnibus per Feature (per-user rates)
effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
i_liwc22	169	291	96	omnibus	9.7306	0.00771	0.0175		0.184479	False
sad_liwc2015	169	291	96	omnibus	9.2763	0.009676	0.0167		0.184479	False
sad_liwc22	169	291	96	omnibus	9.1888	0.010108	0.0166		0.184479	False
i_liwc2015	169	291	96	omnibus	10.0542	0.006558	0.0181		0.184479	False
ppron_liwc2015	169	291	96	omnibus	7.9128	0.019132	0.0143		0.215846	False
netspeak_liwc22	169	291	96	omnibus	7.627	0.022071	0.0137		0.215846	False
netspeak_liwc2015	169	291	96	omnibus	7.4884	0.023654	0.0135		0.215846	False
ppron_liwc22	169	291	96	omnibus	7.6033	0.022334	0.0137		0.215846	False
prep_liwc22	169	291	96	omnibus	6.5633	0.037565	0.0118		0.228523	False
pronoun_liwc2015	169	291	96	omnibus	6.9183	0.031456	0.0125		0.228523	False
pronoun_liwc22	169	291	96	omnibus	6.5734	0.037377	0.0118		0.228523	False
prep_liwc2015	169	291	96	omnibus	6.6142	0.036622	0.0119		0.228523	False
informal_liwc22	169	291	96	omnibus	5.4247	0.066381	0.0098		0.372755	False
informal_liwc2015	169	291	96	omnibus	4.9998	0.082093	0.009		0.428058	False
quant_liwc2015	169	291	96	omnibus	2.8017	0.246382	0.005		0.539319	False
shehe_liwc2015	169	291	96	omnibus	2.7987	0.246763	0.005		0.539319	False
anx_liwc22	169	291	96	omnibus	2.9797	0.225402	0.0054		0.539319	False
quant_liwc22	169	291	96	omnibus	2.8156	0.244684	0.0051		0.539319	False
function_liwc22	169	291	96	omnibus	3.7842	0.150754	0.0068		0.539319	False
interrog_liwc22	169	291	96	omnibus	3.7826	0.150878	0.0068		0.539319	False
shehe_liwc22	169	291	96	omnibus	2.6841	0.261313	0.0048		0.539319	False
ipron_liwc22	169	291	96	omnibus	3.0943	0.212857	0.0056		0.539319	False
compare_liwc22	169	291	96	omnibus	2.7725	0.250015	0.005		0.539319	False
swear_liwc2015	169	291	96	omnibus	3.8183	0.148209	0.0069		0.539319	False
negate_liwc22	169	291	96	omnibus	3.0507	0.217546	0.0055		0.539319	False
filler_liwc22	169	291	96	omnibus	3.1052	0.211693	0.0056		0.539319	False
conj_liwc2015	169	291	96	omnibus	3.5193	0.172101	0.0063		0.539319	False
anx_liwc2015	169	291	96	omnibus	3.258	0.196129	0.0059		0.539319	False
swear_liwc22	169	291	96	omnibus	3.8162	0.148361	0.0069		0.539319	False
compare_liwc2015	169	291	96	omnibus	2.7462	0.253322	0.0049		0.539319	False
filler_liwc2015	169	291	96	omnibus	2.6488	0.265966	0.0048		0.539319	False
conj_liwc22	169	291	96	omnibus	3.4181	0.18104	0.0062		0.539319	False
interrog_liwc2015	169	291	96	omnibus	3.7834	0.150814	0.0068		0.539319	False
ipron_liwc2015	169	291	96	omnibus	3.0614	0.216386	0.0055		0.539319	False
function_liwc2015	169	291	96	omnibus	3.9407	0.139408	0.0071		0.539319	False
negate_liwc2015	169	291	96	omnibus	3.0605	0.216481	0.0055		0.539319	False
focuspresent_liwc22	169	291	96	omnibus	2.3885	0.302927	0.0043		0.574196	False
assent_liwc22	169	291	96	omnibus	2.3749	0.305006	0.0043		0.574196	False
assent_liwc2015	169	291	96	omnibus	2.3634	0.306762	0.0043		0.574196	False
focuspresent_liwc2015	169	291	96	omnibus	2.2269	0.328432	0.004		0.599389	False
anger_liwc22	169	291	96	omnibus	1.9617	0.374993	0.0035		0.622147	False
focuspast_liwc2015	169	291	96	omnibus	2.0522	0.358398	0.0037		0.622147	False
focuspast_liwc22	169	291	96	omnibus	2.0303	0.362346	0.0037		0.622147	False
anger_liwc2015	169	291	96	omnibus	1.9657	0.374237	0.0035		0.622147	False
auxverb_liwc22	169	291	96	omnibus	1.7739	0.411902	0.0032		0.632661	False
they_liwc2015	169	291	96	omnibus	1.7682	0.413077	0.0032		0.632661	False
they_liwc22	169	291	96	omnibus	1.7825	0.410149	0.0032		0.632661	False
auxverb_liwc2015	169	291	96	omnibus	1.7129	0.424663	0.0031		0.632661	False
number_liwc22	169	291	96	omnibus	1.7524	0.416359	0.0032		0.632661	False
focusfuture_liwc2015	169	291	96	omnibus	1.5024	0.47179	0.0027		0.675307	False
focusfuture_liwc22	169	291	96	omnibus	1.5296	0.465421	0.0028		0.675307	False
absolutist_liwc2015	169	291	96	omnibus	1.137	0.56638	0.002		0.736076	False
adverb_liwc2015	169	291	96	omnibus	1.1077	0.574744	0.002		0.736076	False
article_liwc22	169	291	96	omnibus	1.2256	0.541837	0.0022		0.736076	False
article_liwc2015	169	291	96	omnibus	1.1873	0.552302	0.0021		0.736076	False
number_liwc2015	169	291	96	omnibus	1.2578	0.533186	0.0023		0.736076	False
adverb_liwc22	169	291	96	omnibus	1.1419	0.564987	0.0021		0.736076	False
verb_liwc22	169	291	96	omnibus	0.3649	0.833246	0.0007		0.919775	False
affect_liwc2015	169	291	96	omnibus	0.2684	0.874423	0.0005		0.919775	False
negemo_liwc2015	169	291	96	omnibus	0.3289	0.848374	0.0006		0.919775	False
nonflu_liwc2015	169	291	96	omnibus	0.4427	0.801456	0.0008		0.919775	False
posemo_liwc2015	169	291	96	omnibus	0.539	0.763744	0.001		0.919775	False
verb_liwc2015	169	291	96	omnibus	0.2991	0.861096	0.0005		0.919775	False
adj_liwc2015	169	291	96	omnibus	0.3506	0.83919	0.0006		0.919775	False
negemo_liwc22	169	291	96	omnibus	0.2681	0.874566	0.0005		0.919775	False
posemo_liwc22	169	291	96	omnibus	0.5524	0.758655	0.001		0.919775	False
affect_liwc22	169	291	96	omnibus	0.2526	0.881348	0.0005		0.919775	False
nonflu_liwc22	169	291	96	omnibus	0.3327	0.846735	0.0006		0.919775	False
you_liwc22	169	291	96	omnibus	0.2378	0.887914	0.0004		0.919775	False
adj_liwc22	169	291	96	omnibus	0.3453	0.841418	0.0006		0.919775	False
you_liwc2015	169	291	96	omnibus	0.2228	0.894575	0.0004		0.919775	False
we_liwc22	169	291	96	omnibus	0.072	0.96464	0.0001		0.970085	False
we_liwc2015	169	291	96	omnibus	0.0607	0.970085	0.0001		0.970085	False

No features survived omnibus BH correction — no post-hoc tests run

Section 4 — Sensitivity: Single-Period Users Only

Same Design, Restricted to Users Appearing in Exactly One Period
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
netspeak_liwc2015	91.0	182.0	45.0	omnibus	14.547	0.000694	0.0459		0.018267	True
netspeak_liwc22	91.0	182.0	45.0	omnibus	13.863	0.000977	0.0437		0.019286	True
swear_liwc22	91.0	182.0	45.0	omnibus	9.1426	0.010344	0.0288		0.116744	False
informal_liwc22	91.0	182.0	45.0	omnibus	9.2583	0.009763	0.0292		0.116744	False
swear_liwc2015	91.0	182.0	45.0	omnibus	9.1494	0.01031	0.0289		0.116744	False
informal_liwc2015	91.0	182.0	45.0	omnibus	8.5249	0.014088	0.0269		0.139119	False
adj_liwc2015	91.0	182.0	45.0	omnibus	4.8756	0.087352	0.0154		0.462613	False
adj_liwc22	91.0	182.0	45.0	omnibus	4.7972	0.090845	0.0151		0.462613	False
ipron_liwc22	91.0	182.0	45.0	omnibus	5.5501	0.062346	0.0175		0.462613	False
pronoun_liwc2015	91.0	182.0	45.0	omnibus	4.7354	0.093694	0.0149		0.462613	False
ipron_liwc2015	91.0	182.0	45.0	omnibus	5.3258	0.069746	0.0168		0.462613	False
absolutist_liwc2015	91.0	182.0	45.0	omnibus	5.3788	0.067921	0.017		0.462613	False
filler_liwc22	91.0	182.0	45.0	omnibus	4.5851	0.101007	0.0145		0.468054	False
pronoun_liwc22	91.0	182.0	45.0	omnibus	4.4765	0.106645	0.0141		0.468054	False
focusfuture_liwc22	91.0	182.0	45.0	omnibus	3.54	0.170329	0.0112		0.482354	False
i_liwc2015	91.0	182.0	45.0	omnibus	3.7926	0.150121	0.012		0.482354	False
filler_liwc2015	91.0	182.0	45.0	omnibus	3.9493	0.138813	0.0125		0.482354	False
interrog_liwc2015	91.0	182.0	45.0	omnibus	3.4094	0.181828	0.0108		0.482354	False
sad_liwc22	91.0	182.0	45.0	omnibus	3.204	0.20149	0.0101		0.482354	False
i_liwc22	91.0	182.0	45.0	omnibus	3.2506	0.196849	0.0103		0.482354	False
number_liwc22	91.0	182.0	45.0	omnibus	4.2691	0.118299	0.0135		0.482354	False
interrog_liwc22	91.0	182.0	45.0	omnibus	3.4382	0.179228	0.0108		0.482354	False
number_liwc2015	91.0	182.0	45.0	omnibus	3.9936	0.135767	0.0126		0.482354	False
focusfuture_liwc2015	91.0	182.0	45.0	omnibus	3.4815	0.175393	0.011		0.482354	False
article_liwc22	91.0	182.0	45.0	omnibus	3.3392	0.188321	0.0105		0.482354	False
article_liwc2015	91.0	182.0	45.0	omnibus	3.2542	0.196501	0.0103		0.482354	False
sad_liwc2015	91.0	182.0	45.0	omnibus	3.2375	0.198145	0.0102		0.482354	False
we_liwc2015	91.0	182.0	45.0	omnibus	2.8017	0.246393	0.0088		0.486626	False
we_liwc22	91.0	182.0	45.0	omnibus	2.9183	0.232435	0.0092		0.486626	False
assent_liwc2015	91.0	182.0	45.0	omnibus	2.9444	0.229419	0.0093		0.486626	False
ppron_liwc2015	91.0	182.0	45.0	omnibus	2.828	0.243167	0.0089		0.486626	False
assent_liwc22	91.0	182.0	45.0	omnibus	2.9605	0.227579	0.0093		0.486626	False
anx_liwc22	91.0	182.0	45.0	omnibus	3.0622	0.216298	0.0097		0.486626	False
anx_liwc2015	91.0	182.0	45.0	omnibus	2.8333	0.242521	0.0089		0.486626	False
ppron_liwc22	91.0	182.0	45.0	omnibus	2.158	0.339935	0.0068		0.583674	False
focuspast_liwc22	91.0	182.0	45.0	omnibus	2.1154	0.347249	0.0067		0.583674	False
prep_liwc22	91.0	182.0	45.0	omnibus	2.1521	0.340935	0.0068		0.583674	False
focuspast_liwc2015	91.0	182.0	45.0	omnibus	2.228	0.328246	0.007		0.583674	False
nonflu_liwc22	91.0	182.0	45.0	omnibus	2.3179	0.313808	0.0073		0.583674	False
prep_liwc2015	91.0	182.0	45.0	omnibus	2.2921	0.317895	0.0072		0.583674	False
nonflu_liwc2015	91.0	182.0	45.0	omnibus	2.319	0.313639	0.0073		0.583674	False
function_liwc22	91.0	182.0	45.0	omnibus	1.6268	0.443344	0.0051		0.72967	False
function_liwc2015	91.0	182.0	45.0	omnibus	1.487	0.475441	0.0047		0.766527	False
adverb_liwc22	91.0	182.0	45.0	omnibus	1.2132	0.545197	0.0038		0.812653	False
negate_liwc2015	91.0	182.0	45.0	omnibus	1.253	0.534456	0.004		0.812653	False
negate_liwc22	91.0	182.0	45.0	omnibus	1.2251	0.541955	0.0039		0.812653	False
adverb_liwc2015	91.0	182.0	45.0	omnibus	1.2741	0.528858	0.004		0.812653	False
posemo_liwc22	91.0	182.0	45.0	omnibus	0.7431	0.68965	0.0023		0.889588	False
posemo_liwc2015	91.0	182.0	45.0	omnibus	0.727	0.695237	0.0023		0.889588	False
conj_liwc22	91.0	182.0	45.0	omnibus	0.7734	0.67928	0.0024		0.889588	False
conj_liwc2015	91.0	182.0	45.0	omnibus	0.7296	0.694319	0.0023		0.889588	False
quant_liwc2015	91.0	182.0	45.0	omnibus	0.6866	0.709419	0.0022		0.889588	False
anger_liwc22	91.0	182.0	45.0	omnibus	0.8653	0.648791	0.0027		0.889588	False
shehe_liwc22	91.0	182.0	45.0	omnibus	0.8462	0.655022	0.0027		0.889588	False
shehe_liwc2015	91.0	182.0	45.0	omnibus	0.8247	0.662076	0.0026		0.889588	False
anger_liwc2015	91.0	182.0	45.0	omnibus	0.8826	0.643204	0.0028		0.889588	False
quant_liwc22	91.0	182.0	45.0	omnibus	0.7132	0.700047	0.0022		0.889588	False
focuspresent_liwc22	91.0	182.0	45.0	omnibus	0.4046	0.816849	0.0013		0.942002	False
compare_liwc2015	91.0	182.0	45.0	omnibus	0.421	0.810172	0.0013		0.942002	False
compare_liwc22	91.0	182.0	45.0	omnibus	0.4592	0.794866	0.0014		0.942002	False
auxverb_liwc22	91.0	182.0	45.0	omnibus	0.3878	0.823722	0.0012		0.942002	False
auxverb_liwc2015	91.0	182.0	45.0	omnibus	0.3509	0.839096	0.0011		0.942002	False
focuspresent_liwc2015	91.0	182.0	45.0	omnibus	0.333	0.846609	0.0011		0.942002	False
affect_liwc22	91.0	182.0	45.0	omnibus	0.414	0.813021	0.0013		0.942002	False
affect_liwc2015	91.0	182.0	45.0	omnibus	0.4316	0.805897	0.0014		0.942002	False
they_liwc2015	91.0	182.0	45.0	omnibus	0.1776	0.915027	0.0006		0.967971	False
negemo_liwc22	91.0	182.0	45.0	omnibus	0.1562	0.924857	0.0005		0.967971	False
verb_liwc2015	91.0	182.0	45.0	omnibus	0.1164	0.943465	0.0004		0.967971	False
verb_liwc22	91.0	182.0	45.0	omnibus	0.1332	0.935579	0.0004		0.967971	False
they_liwc22	91.0	182.0	45.0	omnibus	0.1954	0.906943	0.0006		0.967971	False
negemo_liwc2015	91.0	182.0	45.0	omnibus	0.1881	0.910217	0.0006		0.967971	False
you_liwc2015	91.0	182.0	45.0	omnibus	0.0052	0.997406	0.0		0.997906	False
you_liwc22	91.0	182.0	45.0	omnibus	0.0042	0.997906	0.0		0.997906	False